



ENTOFARM

INSECT FARMING AS A MODEL FOR CIRCULAR ECONOMY

<https://www.ento.farm/>



MODULE OVERVIEW

This module introduces insect farming as an innovative and sustainable practice within the circular economy. It explores how insects can convert organic waste into valuable products such as protein, animal feed, and fertilizer while reducing environmental impact. By examining insect rearing systems and their benefits compared to traditional livestock farming, learners understand how this approach can help address growing food demand, waste management challenges, and resource sustainability.

TARGET GROUP

Age Range: 12-18 years

Subject Links: STEM, Biology, Sustainability studies, Environmental Science, Civic Education, Economics

LEARNING OBJECTIVES

By the end of this module, learners will:

Knowledge: Define insect farming and the circular economy.
Explain its contribution to sustainability.

Skills: Analyze food systems from a circular economy perspective.

Attitudes & Values: Develop openness to alternative and sustainable food sources.



SCAN ME

Who are we?

About the Project

This module has been developed within the framework of an **Erasmus+ proposal** focusing on **sustainable agriculture, circular economy, and innovative learning** approaches in STEM education.

The project brings together educational institutions, universities, and organizations from **Greece, Bulgaria and North Macedonia** to develop engaging educational materials for secondary school students. Its main objective is to help young people better understand how sustainable practices can address environmental challenges while **encouraging interest in STEM education and career**.

A central element of the project is the EntoFARM educational concept, which introduces students to alternative and sustainable food production systems such as insect farming and other circular resource management practices. Through these examples, students explore how biological processes, technological innovation, and responsible resource use can contribute to more sustainable food systems.

The educational materials produced in the project aim to support teachers in introducing sustainability and circular economy concepts through interactive and interdisciplinary learning activities.



Project Consortium:

- Fraud Line – Greece (Project Coordinator)
- University of Thessaly – Greece
- University of Patras – Greece
- 1st EPAL Volou – Greece
- Professional Agricultural School “Buzema” – Bulgaria
- Professional Forum for Education – Bulgaria
- Association for Internationalization of Education and Science – North Macedonia

The project is co-funded by the European Union. The material of the project EntoFARM reflects only the author's views. The European Commission's support for the production of this material does not constitute an endorsement of the contents which reflects the views only of the authors, and the Commission or the Hellenic National Agency cannot be held responsible for any use which may be made of the information contained therein.

KEY CONCEPTS & THEMATIC CONTENT

Insect Farming:

Insect farming has attracted a lot of scientific and commercial interest during the last decade, as insects are considered a promising alternative source of nutrients that can be exploited for animal feed, as well as for human consumption. And this not unjustifiably, as insects have high nutritional value, are highly efficient in converting feed into insect biomass, can grow well on a huge variety of low-cost organic side-streams and wastes, whereas their production has low land and water requirements and reduced greenhouse gas emissions (Table 1). The EU regulations currently permit and regulate the use of insect-based protein from 8 insects species (Table 2) in aquafeeds, as well as in poultry and pork production (Table 3).

Table 1. Advantages of insects as food and feed

Nutritional value	High-quality protein with a balanced amino acid profile; rich in essential fatty acids, vitamins (B12), and minerals (iron, zinc etc.)
Feed conversion efficiency	Very efficient conversion of feed into biomass compared to livestock (high feed conversion ratio efficiency)
Environmental sustainability	Low greenhouse gas emissions, reduced water use, and smaller land footprint than conventional livestock
Circular economy	Can be reared on organic by-products and food waste, contributing to waste reduction and nutrient recycling
Rapid growth & reproduction	Short life cycles and high reproduction rates enable fast and scalable production

KEY CONCEPTS & THEMATIC CONTENT

Table 1. Advantages of insects as food and feed

Reduced requirements	resource	Require less feed, water, and space compared to cattle, pigs, or poultry
Animal feed applications		Suitable protein source for aquaculture, poultry, pigs, and pet food
Food security		Provides an alternative protein source that can help meet growing global protein demand
Economic potential		Opportunities for new agri-food value chains, rural development, and job creation
Lower disease transmission risk		Lower risk of zoonotic disease transmission compared to traditional livestock (when properly managed)

Table 2. Insect species currently authorized for feed production in the EU

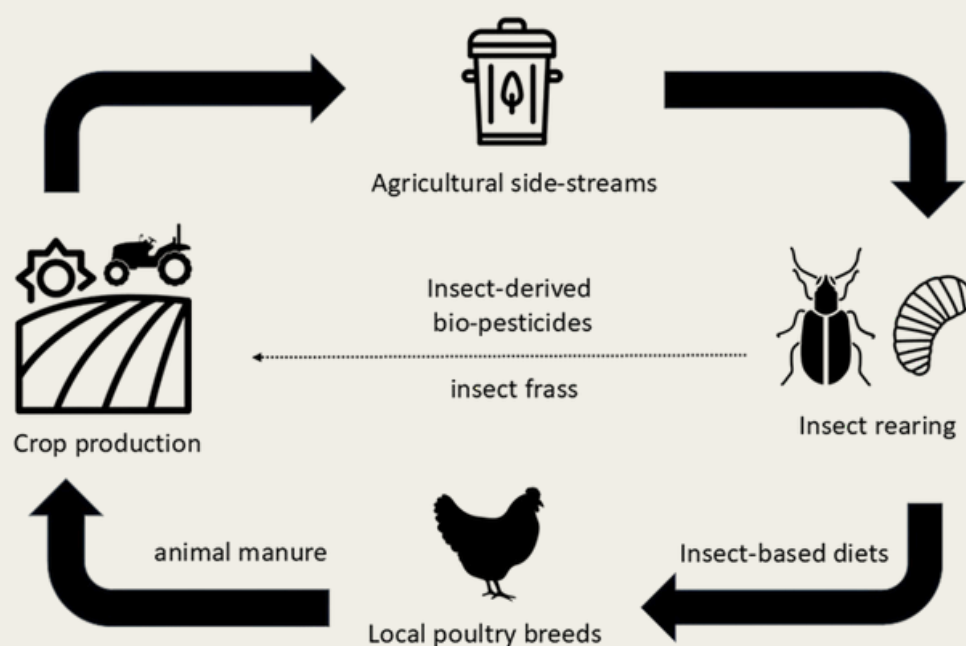
Hermetia illucens	Black soldier fly	Grylloides sigillatus	Banded cricket
Tenebrio molitor	Yellow mealworm	Gryllus assimilis	Field cricket
Alphitobius diaperinus	Lesser mealworm	Locusta migratoria	Migratory locust
Acheta domesticus	House cricket	Bombyx mori	Silkworm

KEY CONCEPTS & THEMATIC CONTENT

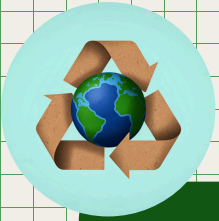
Table 3. Permitted uses of insects as feed in EU

Live insects	Fish, poultry, pigs, pets
Processed insect protein (PAP)	Aquaculture species (permitted since 2021)
Processed insect protein (PAP)	Poultry (permitted since 2021)
Processed insect protein (PAP)	Pigs (permitted since 2021)
Processed insect protein (PAP)	Pets (dogs, cats) (permitted)
Whole insects (dried/frozen)	Pets, zoo, animals (widely permitted)
Insect oil / fat	All farmed animals (permitted)

Below you can see how insect farming is intended to work (Figure 1):



CORE PRINCIPLES OF INSECT FARMING



1. Circular Economy

The circular economy is an economic and production model that aims to minimize waste and make the most efficient use of resources by keeping materials in use for as long as possible. In the context of insect farming, this concept is demonstrated through systems where organic waste streams are reused as inputs for insect rearing, and the outputs (protein, oils, fertilizer) are reintegrated into food, agriculture, or industrial systems. Unlike linear “take-make-dispose” models, insect farming supports circular flows of materials, energy, and nutrients.



2. Bio-waste Valorization

Bio-waste valorization refers to the process of converting organic waste into valuable products instead of disposing of it in landfills or incineration facilities. In insect farming, food waste, agricultural by-products, or organic residues are used as feed for insects, which efficiently transform low-value waste into high-value resources such as protein-rich biomass and organic fertilizer (frass). This process reduces waste volumes, lowers environmental pollution, and adds economic value to materials that would otherwise be discarded.



3. Sustainable Protein

Sustainable protein refers to protein sources that have a lower environmental footprint compared to conventional animal protein. Insects require significantly less land, water, and feed than livestock such as cattle or poultry, while producing high-quality protein suitable for animal feed and, in some contexts, human consumption. Insect farming contributes to more resilient and sustainable food systems by diversifying protein sources and reducing pressure on natural resources.

CORE PRINCIPLES OF INSECT FARMING



4. Resource Efficiency

Resource efficiency is the principle of using fewer natural resources to produce the same or greater value. Insect farming is highly resource-efficient because insects convert feed into body mass very effectively and can thrive on organic side streams rather than dedicated feed crops. This efficiency leads to reduced energy use, lower greenhouse gas emissions, and optimized use of land and water, making insect farming a strong example of efficient biological production systems.



5. Closed-loop Systems

Closed-loop systems are production systems in which outputs, by-products, or waste are reused as inputs, creating minimal or no waste. In insect farming, organic waste feeds insects, insects produce protein and fertilizer, and these outputs are reintegrated into agriculture or food production. This closed-loop approach demonstrates how biological systems can be designed to mimic natural ecosystems, where materials continuously cycle rather than being lost.



6. Life-cycle Thinking

Life-cycle thinking involves evaluating the environmental impacts of a product or system across all stages of its life cycle, from resource extraction to production, use, and disposal. Applying life-cycle thinking to insect farming helps learners understand how insect-based products compare to traditional protein sources in terms of emissions, energy use, and waste generation. This perspective encourages informed decision-making and highlights the broader sustainability benefits of insect farming within food and agricultural systems.

MISCONCEPTIONS TO ADDRESS

Misconception	Clarification
Insects are unsafe as food or feed	Approved insect species are safe and nutritious, and their use is regulated by food and feed safety authorities (e.g. EFSA in the EU).
Insects spread diseases	Farmed insects have a low risk of zoonotic disease transmission, especially compared to conventional livestock.
People will never accept insects	Cultural resistance is changing; insect-based products are increasingly accepted, especially when insects are processed into powders or feeds.
Insects require a lot of resources	Insects are highly efficient, requiring significantly less land, water, and feed than traditional livestock.
Insect farming will replace traditional agriculture	Insect farming complements agriculture by recycling organic waste and providing sustainable feed and fertilizer inputs.
Insect frass is waste	Frass is a valuable by-product used as an organic fertilizer and soil conditioner, contributing to nutrient recycling.
Insect farming is only suitable for large companies	Small-scale and community-level insect farming (including schools) is feasible with proper guidance and infrastructure.

REAL WORLD APPLICATIONS

1. Insects converting food waste into animal feed

Insect farming enables the transformation of food waste and organic by-products into high-quality animal feed. Species such as black soldier fly larvae efficiently consume food scraps from households, supermarkets, and food processing facilities, converting them into protein- and fat-rich biomass. This insect-based feed can be used for poultry, fish, and pets, reducing dependence on conventional feed sources like soy and fishmeal while lowering waste volumes and environmental impact.

2. Insect Frass Used as Organic Fertilizer

Insect frass, a by-product of insect farming consisting of insect waste, shed exoskeletons, and leftover feed, is a nutrient-rich organic fertilizer. When applied to soil, frass improves soil structure, enhances microbial activity, and provides essential nutrients for plant growth. Using insect frass supports sustainable agriculture by reducing reliance on synthetic fertilizers and closing nutrient loops between food production and waste management.

3. Urban Vertical Insect Farms

Urban vertical insect farms are compact, space-efficient production systems designed to operate within cities. These farms use stacked rearing units and controlled environments to produce insects year-round with minimal land and water use. By locating production close to waste sources and consumers, urban insect farms reduce transportation emissions, support local circular economies, and demonstrate innovative solutions for sustainable urban food and feed production.

4. Aquaculture Feed Production

Insect protein is increasingly used as a sustainable alternative to fishmeal in aquaculture feed. Farmed insects provide essential amino acids and nutrients needed for fish growth, while reducing pressure on wild fish stocks used for traditional fishmeal. Incorporating insect-based feed into aquaculture supports more sustainable seafood production and helps address environmental challenges associated with overfishing and marine ecosystem degradation.

5. Sustainable Pet Food Production

Insect-based protein is emerging as a sustainable ingredient in pet food, offering high nutritional value with a lower environmental footprint. Insects require fewer resources to produce than conventional meat proteins and can be raised on organic side streams. In addition, insect protein is often hypoallergenic, making it suitable for pets with food sensitivities. This application highlights how insect farming can contribute to more sustainable consumption patterns beyond human food systems.

CASE STUDY: NASEKOMO

Circular strategy/ Background:

Waste-to-resource and regenerative bioeconomy

Current food systems generate significant quantities of organic waste, while conventional animal feed production depends largely on soy and fishmeal, contributing to deforestation and overexploitation of marine resources.



What the company does and what circular economy principles were applied?

Nasekomo utilizes organic food waste as a feed substrate for insects, primarily black soldier fly larvae. The insects are subsequently processed into protein-rich animal feed and insect oil, while the remaining by-product (frass) is used as a natural fertilizer, thereby returning nutrients to the soil.

Circular economy principles applied:

Repurpose: Converts food waste into high-value products

Recycle (biological loop): Returns nutrients back into the food system

Regenerate: Enhances soil health through the use of organic fertilizer



Sector:

Agri-food systems

Why this example matters?

This case study demonstrates a closed-loop biological system, highlighting how circular economy principles can be implemented at an industrial scale. It directly links circular economy approaches with sustainable agriculture and food security.



Reflection questions:

- What resources does insect farming save compared to traditional feed production?
- How does this approach protect ecosystems?
- Could insect farming work at a local or school scale?

CASE STUDY: FROM FOOD WASTE TO GREEN PROTEIN: A SCHOOL INNOVATION PROJECT

Background/Setting:

A secondary school participates in a sustainability project aiming to reduce food waste and promote circular economy education.

School cafeteria, biology lab, and local community garden.



Scenario Summary

A school launches a pilot project where students collect cafeteria food waste and use it to feed black soldier fly larvae. The larvae are converted into protein feed for chickens, while the remaining frass is used as fertilizer in the school garden, promoting learning about sustainability and circular food systems.



Purpose:

To demonstrate how insect farming can transform organic waste into valuable resources while educating students on circular economy principles.



Reflection questions:

- What sustainability problems does this project solve?
- What risks or challenges could arise?
- How can this model be scaled?
- Would you support this system in your community? Why?

ASSESSMENT & EVALUATION

Quiz: Insect Farming as a Model for Circular Economy

Multiple Choice

1. What is a circular economy?

- a. A system where products are used once and then discarded
- b. A model that keeps resources in use for as long as possible
- c. A system that focuses only on recycling plastics
- d. An economy that does not consider waste management

2. What is bio-waste valorization in insect farming?

- a. Throwing food waste into landfills
- b. Converting organic waste into valuable products such as protein and fertilizer
- c. Using synthetic chemicals to grow insects
- d. Selling leftover food directly to markets

3. Which insect species is commonly used in sustainable farming?

- a. Monarch butterfly
- b. Black soldier fly
- c. Honey bee
- d. Ladybug

4. Which of the following is a benefit of insect protein compared to conventional livestock protein?

- a. Higher greenhouse gas emissions
- b. Lower resource use (land, water, feed)
- c. Requires more energy to produce
- d. Cannot be used in animal feed

5. What is insect frass used for?

- a. Feeding humans directly
- b. Organic fertilizer for plants
- c. Packaging material
- d. A pesticide

6. How do urban vertical insect farms support sustainability?

- a. They produce insects in rural areas only
- b. They reduce transportation emissions and optimize space in cities
- c. They rely on importing large amounts of feed
- d. They increase water and land use compared to conventional farms

7. Life-cycle thinking in insect farming involves:

- a. Only focusing on insect growth stages
- b. Evaluating environmental impacts across all stages of production
- c. Ignoring waste management
- d. Only considering economic costs

8. Which of the following is an example of insect-based sustainable animal feed?

- a. Soybeans and corn only
- b. Fishmeal exclusively
- c. Black soldier fly larvae fed on organic waste
- d. Synthetic protein powder

ASSESSMENT & EVALUATION

Quiz: Insect Farming as a Model for Circular Economy

Multiple Choice

9. Which concept describes keeping outputs and by-products in use within a system?

- a. Open-loop system
- b. Linear economy
- c. Closed-loop system
- d. Single-use system

10. Which of the following best describes the environmental advantage of insect farming?

- a. Uses more land than livestock
- b. Converts organic waste into high-value products efficiently
- c. Requires deforestation for insect feed
- d. Produces higher greenhouse gas emissions than cattle

ACTIVITIES

Activity 1: Design Your Own Circular Insect Farm [EntoFARM]

Duration: 60–90 minutes

Objective: Learners apply module knowledge to design a practical, small-scale insect farming system embedded in a circular economy model.

Activities:

- Short recap of circular economy principles (10 minutes)
- Group work: learners design a circular insect farming model for a chosen context (school, city, farm, business) (30–40 minutes)
- Identification of inputs (waste streams), processes (insect rearing), and outputs (protein, fertilizer, by-products)
- Preparation of a visual model (poster, diagram, or digital canvas)
- Group presentations and peer feedback

Expected Outputs:

- Circular economy system diagram
- Short explanation of environmental and social benefits

Skills Developed:

- Systems thinking
- Collaboration
- Creative problem-solving

ACTIVITIES

Activity 2: Exploring Public Acceptance of Insect Farming

Duration: 60–90 minutes

Objective: Learners investigate societal perceptions, consumer attitudes, and barriers to adopting insect-based foods and feeds and develop strategies to improve public acceptance.

Activities:

- Intro mini-lecture (10 minutes): Overview of cultural, psychological, and market factors influencing acceptance of insect-based products.
- Survey & analysis (20 minutes): Learners design a short survey or interview questions to explore community members' attitudes toward eating insects or using insect-based feed.
- Group discussion (20 minutes): Analyze survey findings and identify key concerns (taste, safety, cultural perceptions, environmental awareness).
- Strategy workshop - awareness task (20–30 minutes): Develop educational or marketing approaches to increase acceptance (e.g., recipe ideas or community events). Design a simple awareness campaign (poster, social media post, or pitch) promoting insect farming.
- Reflection (5–10 minutes): Groups present strategies and discuss lessons learned.

Expected Outputs:

- Summary of public perception insights
- Action plan or prototype awareness campaign to increase acceptance

Skills Developed:

- Research and data analysis
- Empathy and cultural literacy
- Communication and advocacy
- Problem-solving for behavioral change

ADDITIONAL RESOURCES

Links:

- **IPIFF – International Platform of Insects as Food and Feed:** [IPIFF](#) PIFF is an EU non-profit organisation which represents the interests of the insect production sector towards EU policymakers, European stakeholders and citizens. Composed of 56 members, IPIFF promotes the use of insects for human consumption and insect-derived products as a top-tier source of nutrients for animal feed.
- **FlyBox - Insect Farming: How to Get Started and Succeed:** [Insect Farming: How to Get Started and Succeed](#) The FlyBox guide provides an overview of how insect farming can transform organic waste into valuable products such as protein, oil, and fertiliser, offering both environmental and economic benefits. It presents practical steps and best practices for starting and scaling a profitable insect farming operation, covering planning, species choice, setup, and waste-to-resource conversion.
- **TED Talk video: Marcel Dicke: Why not eat insects? (18:34):** [Marcel Dicke: Why not eat insects](#) In his TED Talk, entomologist Marcel Dicke argues that eating insects is a sustainable, nutritious alternative to conventional meat and could help address global food security and environmental challenges as the world's population grows. He encourages overcoming cultural aversions to entomophagy by highlighting insects' ecological benefits and culinary potential.
- **CIBI - Center for Insect Biomanufacturing and Innovation:** [CIBI](#) The Center for Insect Biomanufacturing and Innovation (CIBI) is a first-of-its-kind research center driving breakthrough insect-derived technologies that power innovation across biotechnology, feed, and agricultural industries.

TEACHER REFLECTION AND ADAPTATION NOTES

This module has been designed as a flexible educational resource that can be adapted to different classroom contexts, student backgrounds, and educational levels.

Teachers are encouraged to reflect on how the activities and content can be adjusted to best support the needs of their students.

Reflection Questions for Teachers

Before or after implementing the module, teachers may consider the following questions:

- Did the activities engage students with different interests and learning styles?
- Were all students encouraged to participate actively in discussions and group work?
- Did the module help students understand the connection between science, technology, and sustainability?
- Were examples and explanations accessible to students with different levels of prior knowledge?

Accessibility Considerations

To ensure that the module is accessible to all learners, teachers may consider:

- simplifying technical terminology when necessary
- providing visual aids, diagrams, or short videos
- allowing students to express their understanding through different formats (presentations, posters, discussions, or written reflections)
- offering additional guidance or scaffolding for students who may need more support

Supporting Girls' Engagement in STEM

Research shows that girls are more likely to engage with STEM when learning environments are collaborative and when scientific topics are linked to real-world challenges and social impact.

Teachers can therefore:

- encourage mixed-gender teamwork and equal participation
- highlight female role models in sustainability science and agriculture
- frame STEM challenges as problem-solving activities that benefit society and the environment
- ensure that classroom discussions give equal voice to all students

TEACHER REFLECTION AND ADAPTATION NOTES

Adapting the Module

Teachers are free to adapt the module by:

- adjusting the depth of scientific explanations
- modifying activities to fit available time or resources
- connecting the topic to local environmental or agricultural examples
- integrating the module with subjects such as biology, geography, environmental science, or technology

The goal is to ensure that the module remains relevant, inclusive, and meaningful for students in different educational settings.

Do you have any questions?

Please let us know! We are happy to help.

Contact us at:



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